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| **PI Name:** | Klempner, Mark M.D. | | |
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| **Comments:** | My Official Titles are Associate Provost for Research, Conrad Wesselhoeft Professor of Medicine, and Director, National Emerging Infectious Diseases Laboratory | | |
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| | The goal of this program is to develop a revolutionary non-culture based microbial detection, identification and quantification system that can be used as part of a Smart Medical System for exploratory space travel. Rapid detection and identification of microorganisms are critical to many military and civilian applications ranging from food and water safety monitoring, biological warfare agent detection and to diagnostic microbiology of human and other biological specimens. For long-term exploratory space travel there will be a critical need for a smart medical system to monitor the air and water supply for microbial contaminants and there will also be an intermittent need for assessment of biological specimens from symptomatic astronauts. For research purposes it would also be desirable to determine the effects of long-term space travel on the microbial ecology of environments in the vehicle as well as in its inhabitants (man and animal). Current microbial identification systems are based on the gold standard of in vitro culture or DNA/RNA fingerprinting. Both require considerable sample manipulation, delay in readout, are semiquantitative and subject to | | |

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| **Task Description:** | interfering substances and contamination, and require additional processing to resolve complex mixtures of microorganisms. In this proposal we outline a research plan to develop a novel smart medical system to detect and identify bacteria that builds on our preliminary experience with the use of microsensors for microbial detection. Such a system would not only have direct application to the mission of the NSBRI but should result in a significant transfer of technology for land based applications. Specifically we propose the following tasks Specific Tasks:<br>1) Development of "Fingerprinting" phage display libraries which can detect, identify, quantify and discriminate bacterial species in environmental and biological specimens. This aim seeks to select, characterize, amplify and produce affinity ligand reagents that bind to the surface of microbial species and which are derived from phage displayed combinatorial libraries. 2) Application of phage displayed peptides and antibody fragments in a microarray to the surface of a microsensor to demonstrate the microarray microbial fingerprint response to selected bacterial species using optical readout and electronic MEMS resonator arrays and to characterize the sensitivity and specificity for detecting and discriminating between bacterial species using surface "fingerprints".<br><br>3) Development of algorithms from the microarray response for the real time identification and discrimination of bacterial species |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | Detection and identification of microorganisms in the environment and in biological samples are critical for the success of long-term exploratory space missions such as a manned mission to explore Mars. Microbial contamination of the spacecraft's air and water supply would have disastrous consequences that might be sufficient to result in aborting the mission. Similarly, based on the experience of earth orbit manned missions, it can be anticipated that over the multiyear duration of a Mars mission it will be necessary to assess specimens from the crew for the evaluation and recognition of infectious diseases. Indeed NASA and the NSBRI have identified the need to evaluate environmental and biological specimens for the presence of microorganisms as a priority and one of the existing NSBRI focus areas is "Infection, Immunity and Anemia" under the Direction of Dr. William T. Shearer.1 Since contamination of the environment or significant infection in a crew member will likely result from endogenous human and animal flora that are brought on board at the time of departure it is most critical to have a smart medical system with the capacity to detect bacterial flora. In addition, long term exploratory space travel raises important questions related to changing microbial ecology in the astronauts and in the spacecraft's environmental systems that need to be investigated as part of a comprehensive research program for long term space flight. The innovative bacterial detection and identification system described in this proposal would have substantial application to earth based needs including monitoring of air, water and food contamination as well as for human and animal diagnostic microbiology. Before presentation of the smart medical system to detect bacteria which is the focus of this proposal, below we will briefly review the existing means for bacterial detection and identification and their limitations for long term exploratory space travel. The currently available methods for the detection of bacteria in environmental and biological specimens involve the traditional methods of direct staining for visualization of bacteria, serologic techniques that detect antigens from the bacteria or evidence of prior infection by an antibody response to bacterial antigens, a wide array of culture techniques and the most recent addition of nucleic acid based molecular techniques (e.g. polymerase chain reaction based amplifiction methods). In preface to describing our new bacterial detection and identification system we will briefly review each of these methods and describe their limitations as it applies to long-term exploratory space flight.<br>There are numerous stains that are used to visualize and characterize microorganisms. Bacteria are generally visualized with one of the following direct stains: Gram's stain which distinguishes gram-positive from gram-negative bacteria and provides general morphology of cocci vs rods vs filamentous bacteria vs other (e.g. yeast), acid fast stains which are used to visualize mycobacteria, modified acid fast stains to visualize nocardia species and some parasites (e.g. cryptosporidium and isospora), and fluorescent stains including acridine orange which is useful to see otherwise difficult to stain bacteria (e.g. borrelia, campylobacter species, helicobacter, etc) and fluorescein tagged antibodies which are directed against specific bacteria Except for direct and indirect fluorescent antibody tests which require a fluorescent microscope and a trained reader of the stained specimen, all of these staining methods depend on a large number of organisms to be present in the specimen and are not specific enough to identify the species of bacteria. For example in a urine sample it requires about 100,000 bacteria/ml to detect >5 bacteria per visual field at a magnification of 400X and staining can provide only an indication of whether the organism is gram positive or gram negative. The use of bacterial antigen detection tests is limited to a few species under circumstances that have unlikely applicability to the exploratory space travel program. For example there are bacterial antigen detection tests which are based on agglutination and are generally used only for cerebrospinal fluid samples to detect Streptococcus pneumoniae, Hemophilus influenzae type b, Neisseria meningitidis, and Group B streptococci in neonates. Enzyme linked tests are available for detecting Group A streptococci in throat swabs and Legionella pneumophila serotype 1 in sputum and urine specimens. Other antigen agglutination and enzyme immunoassays are available for some fungi, viruses and parasites.<br><br>Identification and quantification of bacteria by in vitro culture and biochemical methods are the "gold standard".2,3 Numerous agar and broth based media are used to cultivate bacteria depending on the source of the specimen and the spectrum of bacteria that are anticipated. General media include sheep blood agar, chocolate agar, eosin-methylene blue (EMB) or MacConkey agar, phenylethyl alcohol agar, thioglycollate broth media, and brain heart infusion broth media. Special media and culture conditions are required for the isolation of specific bacteria. For example anaerobic bacteria are cultivated under anaerobic conditions on kanamycin-vancomycin laked blood agar (Prevotella and Bacteroides species), BSK for Borrelia species, BCYE agar for Legionella species, PC agar for Burholderia cepacia, Tinsdale agar for Corynebacterium diptheriae, and over 25 others. Once the bacteria are grown individual colonies are amplified to obtain pure cultures and subjected to multiple biochemical tests and reactions for species identification. While much of these latter steps are currently automated it would be impractical to have an on board exploratory spacecraft full diagnostic microbiology laboratory, the expertise to run such a facility, and the ability to renew reagents over the duration of the mission.<br><br>Molecular based techniques are growing in importance for diagnostic microbiology.(reviewed in 4) They can be used for organism detection and identification, antimicrobial drug resistance testing and new organism detection. Once organisms have been grown in culture specific identification can be done using DNA probes that hybridize with bacterial ribosomal RNA. The currently available DNA probes are labeled such that hybridized probes are detected by chemiluminescence. This method is being used for the identification of only a few pathogens (e.g. sexually transmitted |

pathogens such as N. gonorrhoeae and C. trachomatis) as well as confirmation of species identification for acid fast bacilli, etc. Polymerase chain reaction (PCR) amplification techniques to detect and identify microorganisms have been under development in many university research laboratories for over 10 years but have been slow to move to the routine diagnostic microbiology laboratory. The exquisite sensitivity of PCR for detecting very few bacteria is both its major advantage and its greatest drawback. While the identification of a very small number of organisms is very helpful in some specimens (e.g. CSF) or for some bacteria (e.g. M. tuberculosis) in many biologic specimens (e.g. urine, sputum, water, air, etc) the number of organisms is critical to judge the significance of the finding. Quantitative PCR has been extremely useful for monitoring HIV infection but standards for bacterial quantification are lacking. Similarly, species specific primers have been designed and validated but the large number of such primers that need to be available for diagnostic microbiology and the problems with false positive reactions, contamination of equipment, and inhibitors of the PCR reaction in biological specimens all present challenges for routine use of this method in the diagnostic microbiology laboratory or for use in long term exploratory space travel. Because primers for antimicrobial drug resistance genes have been identified one of the most promising uses of PCR will be in the prediction of antimicrobial agent sensitivity testing. PCR has also provided a major advance in our ability to detect organisms that cause diseases that were previously considered idiopathic such as in Whipple's disease.

The NSBRI has identified the importance of monitoring the space environment and the astronauts for microorganisms during long duration space travel. Within the Immunology, Infection and Anemia research focus area led by Dr. William T. Shearer, George Fox, Ph.D., of the University of Houston, has a small program for the design of DNA probes that could be useful for monitoring the on board water system.5 It is not intended as a human pathogen detection system and demonstration probes in a microarray, luminescent readout are being developed for coliform bacteria. While there is potential overlap with this proposal in this particular area, PCR based technologies will have at least the same diagnostic microbiology drawbacks in the space environment as they do in the Earth based diagnostic microbiology laboratory. In discussions with Dr. Fox there could be opportunities for synergy between these projects as both proposers are interested in collaboration. The research questions that are being posed by these teams also differ in that Dr. Fox will focus on bacterial gene expression in microgravity and this proposal will investigate the longitudinal changes in the microbial ecology in the astronauts and the spacecraft environmental systems during long-term space travel.

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